CAPSTONE PHASE 1 · CLASSICAL MACHINE LEARNING



ONE OF EVERY CLASS · 28 x 28

How Far Can Classical Machine Learning Go on Fashion-MNIST?

A leak-free benchmark of raw pixels vs HOG features, with error analysis.

TEAM

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02 · WHAT THIS TALK ANSWERS

Auto-tag clothes with classical ML, before we reach for deep learning.

Everything in this talk answers one of these four questions. We return to each with a number at the end.

RQ1

How accurate is classical ML on raw pixels?

Six models, same split, no Decision on test set.

RQ2

Do HOG (edge-direction) features beat raw pixels, and for whom?

Linear models may need shape, trees may not.

RQ3

Which model gives the best accuracy / speed / size trade-off?

Honest numbers, including inference time and model footprint.

RQ4

Which classes get confused, and why?

We use per-class recall, the confusion matrix and ambiguous shirts.

Twelve experiments, all reproducible in under two minutes from a fresh clone.

TEST ACCURACY

91.08%

CHOSEN ON VALIDATION, NEVER ON TEST

BEST MODEL

RBF SVM on HOG features

VALIDATION

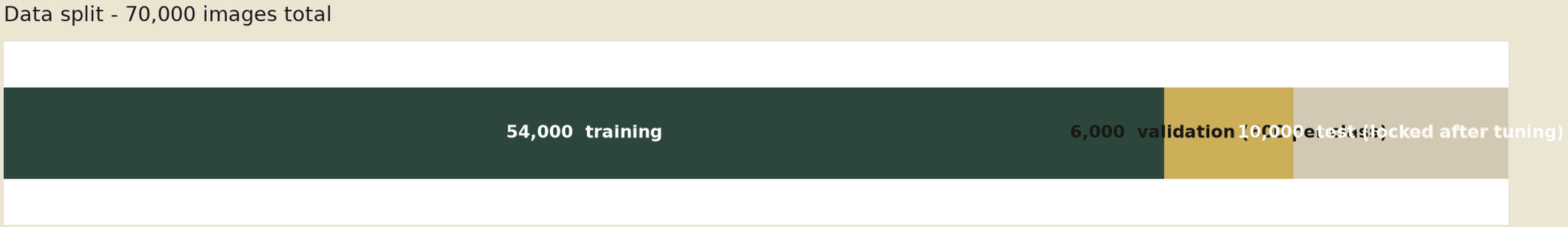
91.57%

HARDEST CLASS

Shirt, mostly confused with T-shirt, Pullover, Coat

PART 1 · DATA PIPELINE

The dataset and a fixed, honest split.



THE DATASET

Fashion-MNIST.
70,000 grayscale 28x28 images,
10 balanced clothing classes.

Verified by MD5, sanity-checked:
0 duplicates, 0 missing values.

THE SPLIT

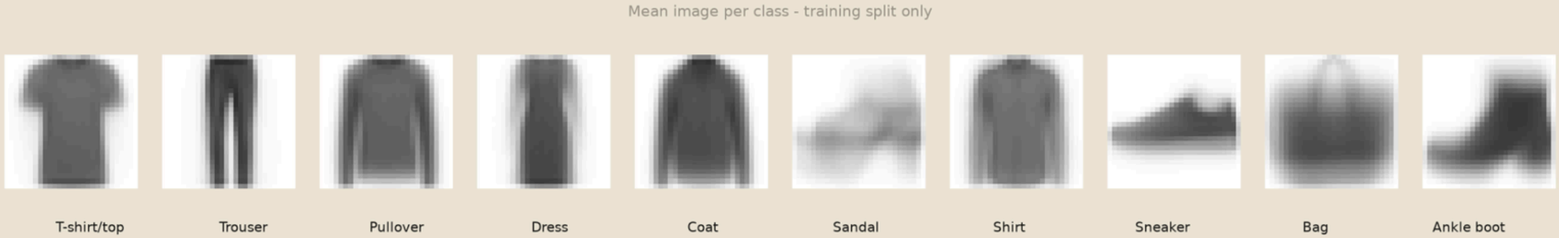
54,000 train · 6,000 validation
(600 per class · seed 42)
· 10,000 test.

Validation indices committed to the repo,
so every team member uses the identical set.

ONE RULE

Train fits, validation decides, test is opened once per experiment at the very end.

Before training a single model.



MEAN IMAGE PER CLASS · TRAINING SPLIT ONLY



T-SNE · 4,000 TRAINING SAMPLES · 10 CLASSES

THE FOUR TOPS COLLIDE

T-shirt, Pullover, Coat, Shirt have near-identical mean images.

Cosine similarity of class means: 0.95 - 0.99.

T-SNE CONFIRMS IT

Pullover, Coat, Shirt form one mixed cloud.

Trouser and Bag form clean, separate clusters.

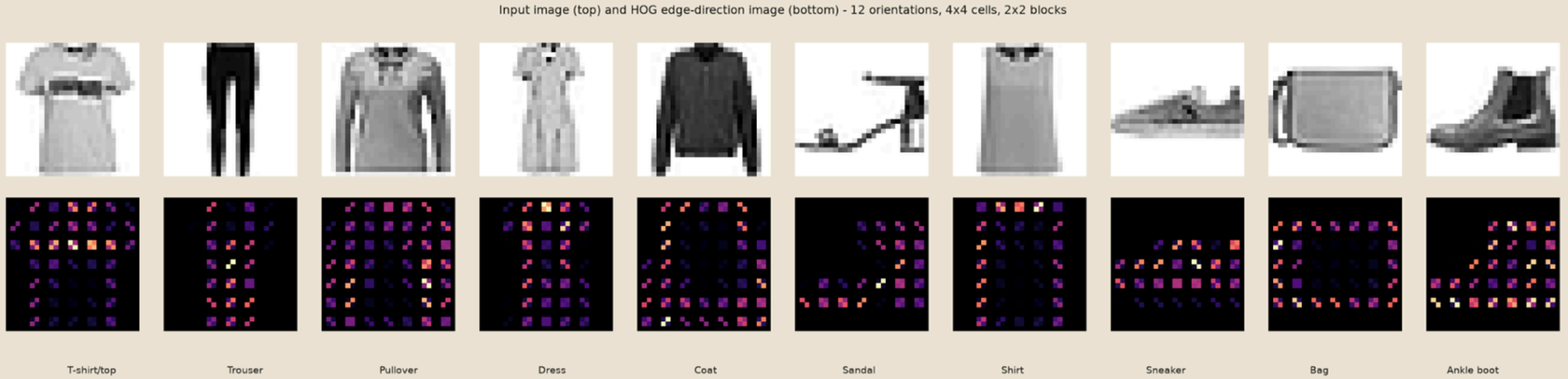
PIXELS ARE REDUNDANT

84 of 784 PCA components keep 90% of the variance.

PREDICTION BEFORE ANY MODEL

Shirt will be the hardest class.

Two ways to describe an image.



RAW PIXELS

784

Flatten 28 x 28 into 784 brightness values.

Simple, but a 2-pixel shift changes everything.

HOG FEATURES

1,728

Edge-direction histograms in small cells, robust to brightness.

12 orientations · 4x4 cells · 2x2 blocks.

RULE FOR CHOOSING SETTINGS

Highest validation accuracy; within 0.3 points prefer the smaller vector.

12-CONFIGURATION SWEEP

Small cells mattered most: +4 to +8 validation points.

EXTRACTION COST

~0.22s

per 1,000 images.

One harness, no leakage.

STEP 1
Train
fit on 54k

→

STEP 2
Validation
choose settings

→

STEP 3
Refit on 60k
chosen settings

→

STEP 4
Test
opened once

→

STEP 5
Save JSON
+ audit log row

ONE DOOR TO TEST

Test set is read by exactly one function.

Every use is written to an audit log: **12 rows for 12 experiments**.

SCALING STAYS IN THE PIPELINE

Every scaler fits only on training rows.

Validation and test are transformed only.

ONE EVALUATION FUNCTION

Accuracy · macro F1 · per-class recall · confusion matrix · time · model size.

Same code, same shape, every model, every run.

> **Leakage is the most common way ML coursework goes wrong. We made it structurally hard.**
>
> The only door to the test set is one function, and it leaves a record every time it is opened.

Baseline, Logistic Regression, k-NN.

DUMMY FLOOR

Always predicts one class.

10.00%

Every trained model must beat this.

LOGISTIC REGRESSION · BEST C = 0.01

PIXELS

84.72%



300 px

+4.14 POINTS WITH HOG

HOG

88.86%



300 px

HOG gives a linear model the shape information it cannot build from raw pixels on its own.

K-NN · K = 3, DISTANCE-WEIGHTED

PIXELS

86.11%

HOG

86.31%

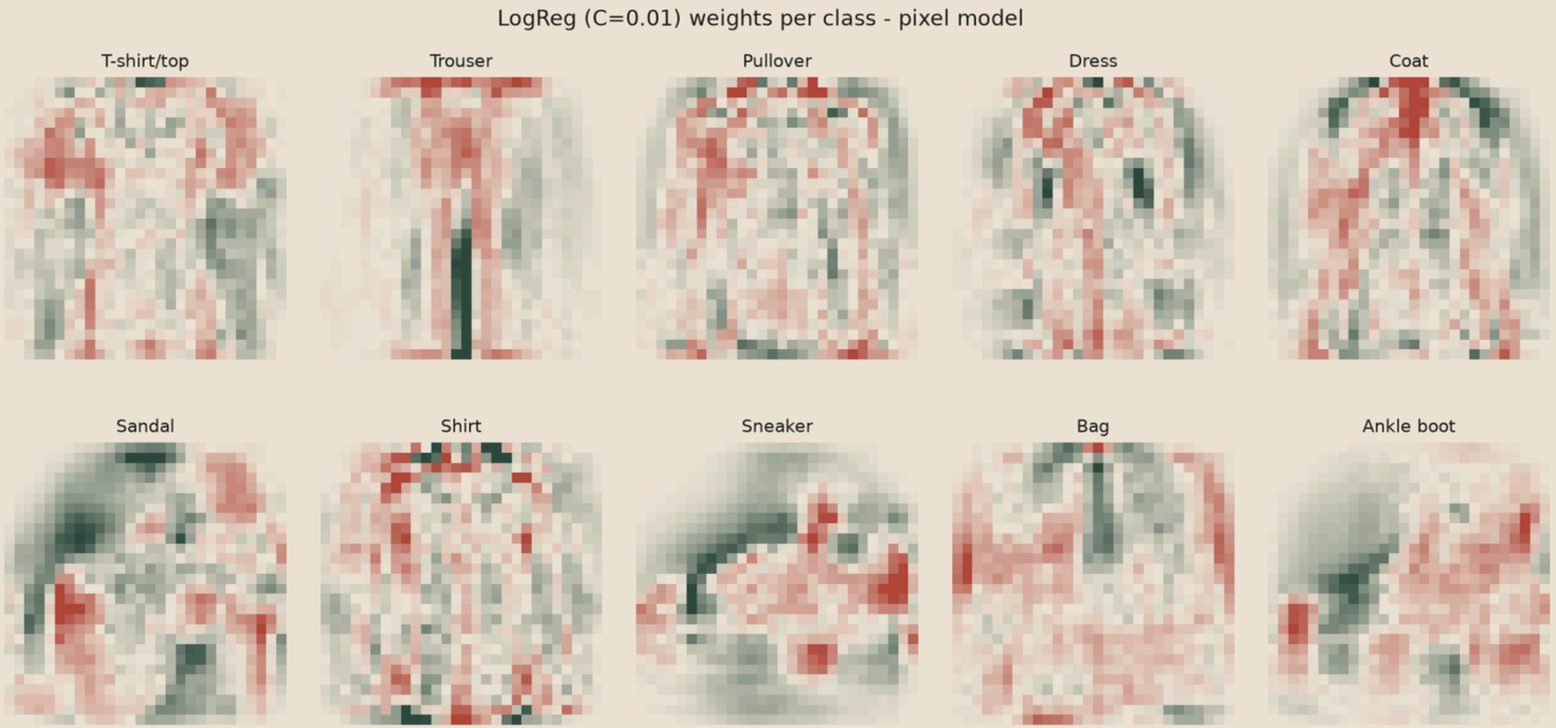
HOG barely helps. Stores all 60k images (189 to 415 MB); 0.7 - 6.9 s per 1,000 to predict.

THEORY

HOG is a hand-engineered fix for one specific blind spot: linear models have no shape prior. k-NN already compares whole images, so swapping the description barely changes its neighbours.

Caveat: best C = 0.01 sits at the edge of the grid we tried. More on this when we look at the linear weights.

What a linear model looks at.



WHAT TO LOOK FOR

TROUSER

A strong negative strip between the legs.

SHOES

Weight lives on the sole and ankle region.

THE FOUR TOPS

Share broad torso and sleeve regions, which is why a linear model struggles with them.

STATED LIMITATION

Best C sits at the edge of the grid we tried (0.01). The true optimum may be a bit lower. We did not widen the grid afterwards, because that would have meant re-tuning and re-testing.

Random Forest, then Linear SVM.

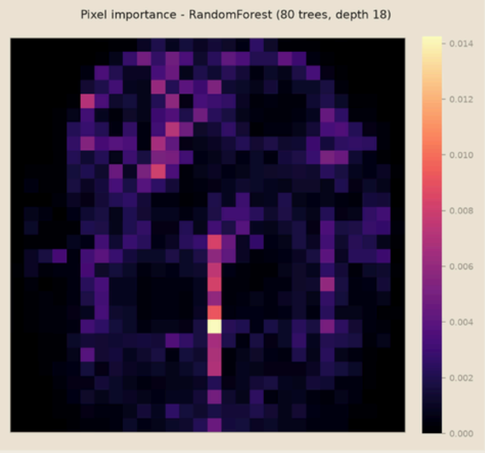
RANDOM FOREST · 300 TREES · DEPTH 20

PIXELS

87.56%

HOG

87.03%



PIXEL IMPORTANCE HEAT MAP

HOG does **not** help the forest. Each split looks at one feature, and one pixel says more than one HOG bin.

LINEAR SVM · BEST C = 0.01

PIXELS

83.78%

HOG

88.98%

BIGGEST HOG GAIN

+5.20

MODEL SIZE

181 KB

Linear SVM on HOG is tiny but slow to tune: each fit on 54k took 105 to 924 s, growing with C.

The most accurate model, and the comparison almost broken it.

RBF SVM · PIXELS

C = 10 · KERNEL RBF · GAMMA = SCALE

89.70%

RBF SVM · HOG

C = 10 · KERNEL RBF · GAMMA = SCALE

TEST ACCURACY

91.08%

THE PROBLEM WE FOUND

Too slow to tune on 54k images. The 9-setting grid was searched on a 10k stratified subset, then refit on 60k.

Side effect: its validation score came from 10k training images while every other model used 54k. So it was unfairly low.

THE FIX, COMMITTED BEFORE LOOKING

Refit the already-chosen settings on 54k and score on validation. No new search. No test.

FAIR VALIDATION RANKING

RBF · HOG	0.9157
RBF · pixels	0.9082
LogReg · HOG	0.8927

Without this fix we would have picked Logistic Regression as "best by validation" for the wrong reason. We fixed the comparison, not the rule, and we committed the plan before seeing the result.

All 12 experiments, on the same split.

MODEL	PIXELS · TEST	HOG · TEST
Dummy baseline	10.00%	10.00%
Logistic Regression	84.72%	88.86%
k-NN	86.11%	86.31%
Random Forest	87.56%	87.03%
Linear SVM	83.78%	88.98%
RBF SVM	**89.70%**	**91.08%**

BEST BY VALIDATION = BEST ON TEST

RBF SVM on HOG

The same model leads every ranking we report.

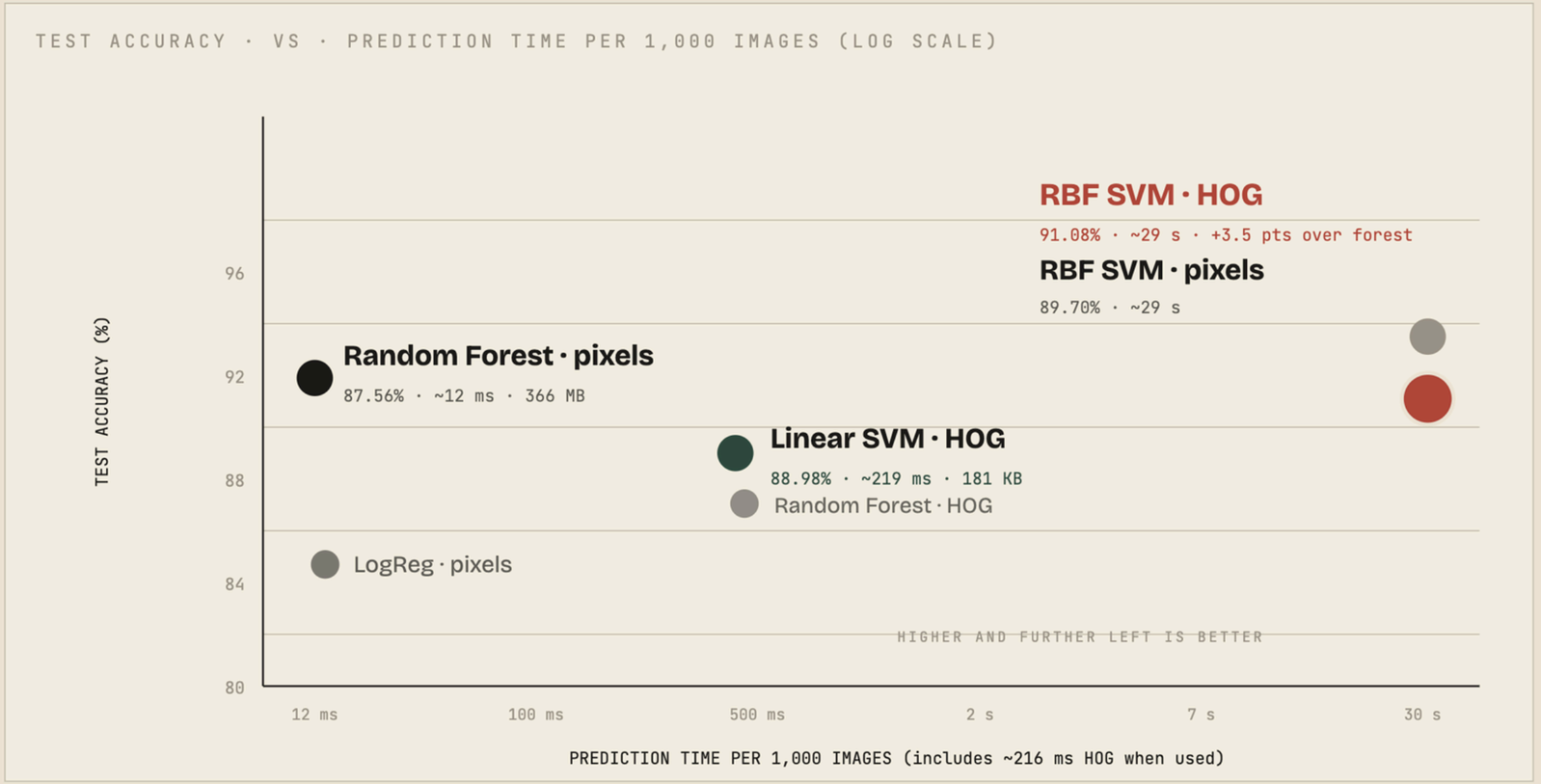
5-FOLD CV (10K SUBSET)

0.8775

± 0.0054 · the model is stable.

RQ1 89.70% from raw pixels. **RQ2** HOG helps linear models a lot, the RBF SVM a little, and not trees or k-NN.

Three models, three places to land.



FASTEST ABOVE 87%

Random Forest · pixels

87.56% · ~12 ms per 1k · 366 MB.

BEST ACCURACY FOR ITS SIZE

Linear SVM · HOG

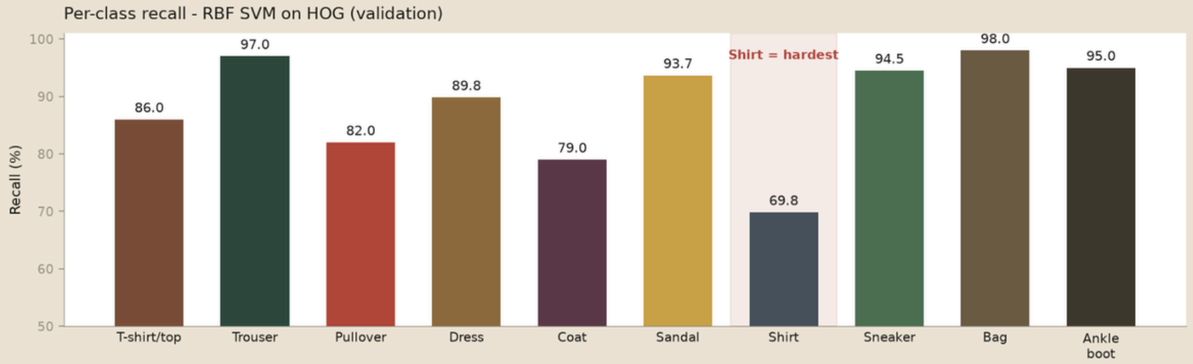
88.98% · 181 KB · ~219 ms per 1k (mostly HOG).

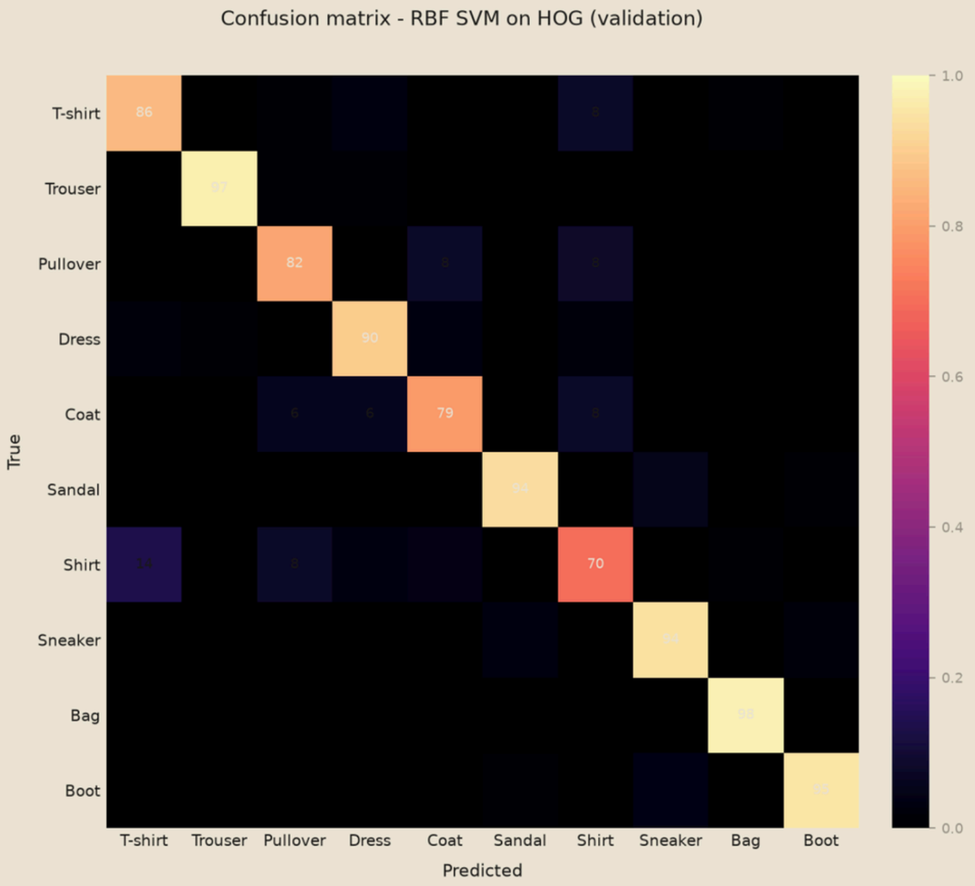
MOST ACCURATE

RBF SVM · HOG

91.08% · ~29 s per 1k · ~2,400x slower than the forest.

Where the models fail.





PATTERN, NOT NOISE

Shirt is the hardest class for all 10 trained models, recall 50.9% to 75.2%.

Mistakes among the four tops are 56-66% of every model's errors — exactly the group the EDA flagged.

At 28 x 28, collars, buttons and cuffs cover only a few pixels.

Same mistakes, different features.

BOTH RIGHT

8,593

Both RBF pixels and RBF HOG agree.

ONLY HOG RIGHT

515

HOG fixed errors from the pixel model.

ONLY PIXELS RIGHT

377

HOG introduced these errors.

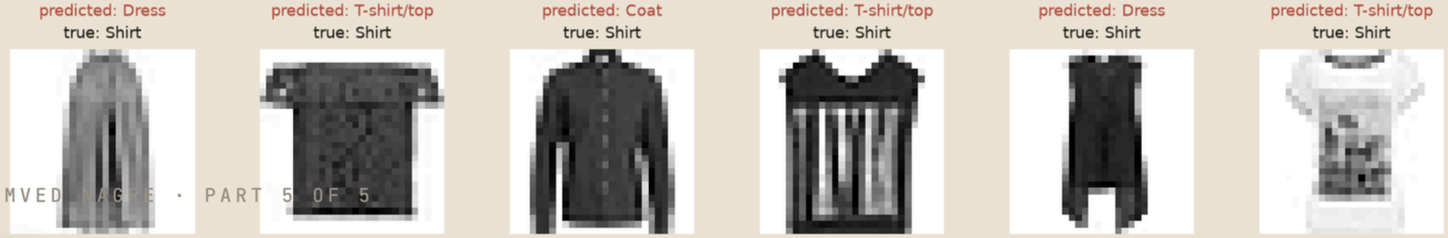
BOTH WRONG · 515 IMAGES

391 of those 515 get the same wrong label from both models. 397 are upper-body garments.

Net effect: HOG fixes 515 of the pixel model's errors and adds 377 new ones. Net gain of 138 images = the 1.38-point test gap.

SOME OF THE SHIRTS THE BEST MODEL MIS-CLASSIFIED

Shirts the RBF SVM mis-classified on validation



SO WHAT

When two very different models make the same mistake, the image is probably ambiguous.

Some of these shirts would fool a person too. Handcrafted features have hit their ceiling here.

Conclusions, and the road to Phase 2.

CLASSICAL CEILING

91.08%

Chosen with zero test-set decisions.

BIGGEST HOG WIN

Linear

+5.20 points for Linear SVM, +4.14 for LogReg. The most accurate model is also the slowest.

LIMITATIONS, NAMED UP FRONT

Best settings at grid edges · two laptop timings · val slightly optimistic.

Test is the honest number, even when it is below validation.

PHASE 1 · DONE

RBF SVM on HOG

91.08% on test.

Same split, same harness, all 12 experiments reproducible in under two minutes from a fresh clone.

THE FOUR TOPS



PHASE 2 · HYPOTHESIS

A small CNN on the same split and the same harness.

Learned filters should catch collars and buttons, and fix the four-top confusion at the core of this benchmark.

Everything is reproducible. Code, split, results, figures and a 32-entry decision log on GitHub.

Each presenter can answer questions on their own part. Thank you.